

IEOR 4404: Simulation

Lecture 7: Monte Carlo to Approximate Integrals

Computing Integrals

- Suppose we want to compute

$$\int_0^1 g(x) dx$$

Linking "Integration" to "Expectation of cont. var."

$$\begin{aligned} E[g(X)] &= (\text{Discrete}) \sum_{x \in \mathcal{X}} g(x) \cdot P(X) \\ &= (\text{Cont.}) \int g(x) \cdot f(x) dx \end{aligned}$$

- Instead of numerical integration, we can use Monte Carlo by noting:

$$E[g(X)] = \int_0^1 g(x) f(x) dx$$

for some density f , where in this case f is the uniform(0,1) density

Computing Integrals

Procedure to approximate the integral:

- Generate $U \sim Unif(0,1)$ for n times, say they are $U_1, \dots, U_n$
- Output

$$\frac{g(U_1) + \dots + g(U_n)}{n}$$

as an approximation

- This approximation is subject to Monte Carlo error, and the output above is called a point estimate (more to come later)

Computing Integrals

Estimate $\int_0^1 (1 - x^2)^{\frac{3}{2}} dx$

Using previous notation, $g(u) = (1 - u^2)^{\frac{3}{2}}$

$$\Rightarrow \int_0^1 (1 - x^2)^{\frac{3}{2}} \cdot \underbrace{1}_{\sim f(x)} dx = \mathbb{E}_f[(1 - u^2)^{\frac{3}{2}}] \text{ where } U \sim \text{Unif}(0, 1)$$

Procedure:

- Generate $u_1, u_2, \dots, u_n \sim \text{Unif}(0, 1)$
- pt. est. $\frac{g(u_1) + \dots + g(u_n)}{n}$

Computing Integrals

Estimate $\int_0^1 (1 - x^2)^{\frac{3}{2}} dx$

Procedure:

- Generate $U \sim \text{Unif}(0,1)$ for n times, say they are $U_1, \dots, U_n$
- Output

$$\frac{g(U_1) + \dots + g(U_n)}{n}$$

where $g(u) = (1 - u^2)^{\frac{3}{2}}$

Computing Integrals with More General Limits

Suppose now we want to compute

$$\int_a^b g(x) dx$$

Thoughts ?

Immediate idea $\Rightarrow$ - Generate $u_1, \dots, u_n \sim \text{Unif}(a, b)$
- pt. est. $\frac{g(u_1) + \dots + g(u_n)}{n}$

Is this correct ? $\Rightarrow$ NO !!

Computing Integrals with More General Limits

Suppose now we want to compute

$$\int_a^b g(x)dx$$

Two approaches:

- Approach 1: Generate $U \sim \text{Unif}(a, b)$
- Approach 2: Change of variable

Computing Integrals with More General Limits

Approach 1: Write

$$\int_a^b g(x) dx = (b - a) E[g(U)]$$

where $U \sim \text{Unif}(a, b)$

$$\begin{aligned} \int_a^b g(x) dx &= \int_a^b g(x) \cdot \overset{f(x) \text{ for } x \sim \text{Unif}(a, b)}{\underbrace{\frac{1}{b-a}}}_{dx} \cdot (b-a) \\ &= \underline{\underline{(b-a)}} \cdot \mathbb{E}[g(U)] \quad \text{where } U \sim \text{Unif}(a, b) \end{aligned}$$

Computing Integrals with More General Limits

Approach 1: Write

$$\int_a^b g(x)dx = (b - a)E[g(U)]$$

where $U \sim \text{Unif}(a, b)$

Procedure:

- Generate $U \sim \text{Unif}(a, b)$ for n times, for example using ITM where $U = a + (b - a)V$ for $V \sim \text{Unif}(0,1)$. Say these realizations are $U_1, \dots, U_n$
- Output

$$(b - a) \frac{g(U_1) + \dots + g(U_n)}{n}$$

Computing Integrals with More General Limits

Approach 2: By a change of variable $x = a + (b - a)y$, or $y = \frac{x-a}{b-a}$,

$$\int_a^b g(x)dx = \int_0^1 g(a + (b - a)y)(b - a)dy$$

and then apply simulation to compute $E[\tilde{g}(U)]$ where

$$\tilde{g}(u) = g(a + (b - a)u)(b - a)$$

and $U \sim \text{Unif}(0,1)$

Looks complicated, but hints toward general $\int_a^b$ setting

$$\begin{aligned} & \begin{cases} x=0 \Rightarrow y=a \\ x=1 \Rightarrow y=b \end{cases} \\ \Rightarrow \text{Change of var.} \Rightarrow x = a + (b-a)y & \Rightarrow dx = (b-a)dy \end{aligned}$$

$$\text{Hence } \int_a^b g(x)dx = \int_0^1 g(a + (b-a)y)(b-a)dy$$

which returns to $\int_0^1$ case with $\tilde{g}(u) = g(a + (b-a)u) \cdot (b-a)$

Computing Integrals with More General Limits

Approach 2: By a change of variable $x = a + (b - a)y$, or $y = \frac{x-a}{b-a}$,

$$\int_a^b g(x)dx = \int_0^1 g(a + (b - a)y)(b - a)dy$$

and then apply simulation to compute $E[\tilde{g}(U)]$ where

$$\tilde{g}(u) = g(a + (b - a)u)(b - a)$$

and $U \sim Unif(0,1)$

Procedure:

- Generate $U \sim Unif(0,1)$ for n times. Say they are $U_1, \dots, U_n$
- Output

$$\frac{\tilde{g}(U_1) + \dots + \tilde{g}(U_n)}{n}$$

Computing Integrals with More General Limits

Estimate $\int_1^3 (1 - x^2)^{\frac{3}{2}} dx$

Approach 1 $\int_1^3 (1 - x^2)^{\frac{3}{2}} = (3-1) \cdot \mathbb{E}[(1 - u^2)^{\frac{3}{2}}] = 2 \cdot \mathbb{E}[g(u)]$
where $g(u) = (1 - u^2)^{\frac{3}{2}}, u \sim \text{Unif}(1, 3)$

Procedure :

- Generate $u_1, u_2, \dots, u_n \sim \text{Unif}(1, 3)$
- pnt. est. $2 \cdot \left[\frac{g(u_1) + \dots + g(u_n)}{n} \right]$

Approach 2 Plug-in $\tilde{g} \Rightarrow \int_1^3 (1 - x^2)^{\frac{3}{2}} dx = \int_0^1 [1 - (1+2y)^2]^{\frac{3}{2}} \cdot 2 dy$
where $\tilde{g}(u) = 2 \cdot [1 - (1+2u)^2]^{\frac{3}{2}}$

Procedure :

- Generate $u_1, u_2, \dots, u_n \sim \text{Unif}(0, 1)$
- pnt. est. $\frac{\tilde{g}(u_1) + \dots + \tilde{g}(u_n)}{n}$

Computing Integrals with Unbounded Limits

Suppose now we want to compute

$$\int_0^{\infty} g(x) dx$$

Need to think more, since Unif $f(x) = c$ over finite region,
yet any $c > 0$ over $[0, \infty]$ will lead to invalid pdf.

Computing Integrals with Unbounded Limits

Suppose now we want to compute

$$\int_0^{\infty} g(x) dx$$

Two approaches:

- Approach 1: Identify the integral as

$$\int_0^{\infty} h(x) f(x) dx$$

for some density f supported on $(0, \infty)$

- Approach 2: Change of variable to make the range from 0 to 1 again, e.g., $Y = 1/(1 + X)$

Computing Integrals with Unbounded Limits

Approach 1: Identify

$$\int_0^{\infty} g(x) dx = \int_0^{\infty} h(x) f(x) dx = E[h(X)]$$

where f is a density on $[0, \infty)$ and the expectation is taken w.r.t.
 $X \sim f$

High-Level idea:

Weighted Average $\Rightarrow$ X 's weight determined
by pdf $f(X)$.

Computing Integrals with Unbounded Limits

Approach 1: Identify

$$\int_0^{\infty} g(x)dx = \int_0^{\infty} h(x)f(x)dx = E[h(X)]$$

where f is a density on $[0, \infty)$ and the expectation is taken w.r.t. $X \sim f$

Procedure:

- Generate $X \sim f$ for n times. Say they are $X_1, \dots, X_n$
- Output

$$\frac{h(U_1) + \dots + h(U_n)}{n}$$

Computing Integrals with Unbounded Limits

Approach 2: By a change of variable $y = \frac{1}{1+x}$,

$$\int_0^{\infty} g(x) dx = \int_0^1 g\left(\frac{1-y}{y}\right) \frac{1}{y^2} dy$$

and then apply simulation to compute $E[\tilde{g}(U)]$ where

$$\tilde{g}(u) = g\left(\frac{1-u}{u}\right) \frac{1}{u^2}$$

and $U \sim \text{Unif}(0,1)$

$$x = \frac{1-y}{y}$$

Derivation :

$$y = \frac{1}{1+x} \Rightarrow \begin{cases} x=0 \Rightarrow y=1 \\ x \rightarrow \infty \Rightarrow y \rightarrow 0 \end{cases} \Rightarrow dy = -\frac{1}{(1+x)^2} dx$$

$$\Rightarrow -\frac{1}{y^2} dy = dx$$

$$\text{hence } \int_0^{\infty} g(x) dx = \int_1^0 g\left(\frac{1-y}{y}\right) \left(-\frac{1}{y^2}\right) dy$$

$$= \int_0^1 g\left(\frac{1-y}{y}\right) \frac{1}{y^2} dy$$

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Computing Integrals with Unbounded Limits

Approach 2: By a change of variable $y = \frac{1}{1+x}$,

$$\int_0^{\infty} g(x) dx = \int_0^1 g\left(\frac{1-y}{y}\right) \frac{1}{y^2} dy$$

and then apply simulation to compute $E[\tilde{g}(U)]$ where

$$\tilde{g}(u) = g\left(\frac{1-u}{u}\right) \frac{1}{u^2}$$

and $U \sim Unif(0,1)$

Procedure:

- Generate $U \sim Unif(0,1)$ for n times. Say they are $U_1, \dots, U_n$
- Output

$$\frac{\tilde{g}(U_1) + \dots + \tilde{g}(U_n)}{n}$$

Computing Integrals with Unbounded Limits

Estimate $\int_0^{\infty} e^{-x^2} dx$

No obvious $f(x) \cdot h(x)$ decomposition, hence Approach 2

$$\int_0^{\infty} e^{-x^2} dx = \int_0^1 e^{-\left(\frac{1-y}{y}\right)^2} \frac{1}{y^2} dy = \mathbb{E}[\tilde{g}(U)]$$

$$\text{where } \tilde{g}(u) = e^{-\left(\frac{1-u}{u}\right)^2} \cdot \frac{1}{u^2}, \quad U \sim \text{Unif}(0, 1)$$

Procedure • Generate $U_1, \dots, U_n \sim \text{Unif}(0, 1)$

$$\bullet \text{ pt. est. } \frac{\tilde{g}(U_1) + \dots + \tilde{g}(U_n)}{n}$$

Computing Integrals with Unbounded Limits

Estimate $\int_0^{\infty} e^{-x} \sin x \, dx$

Clear decomposition $\Rightarrow h(x) = \sin x$, $f(x) = e^{-x}$

hence adopt Approach 1

Procedure :

- Generate $X_1, \dots, X_n \sim \text{Exp}(1)$ via ITM.
- pnt. est. $\frac{\sin(X_1) + \dots + \sin(X_n)}{n}$

Computing Multivariate Integrals

- Suppose now we want to compute a multivariate integral

$$\int_0^1 \cdots \int_0^1 g(x_1, \dots, x_k) dx_1 \cdots dx_k$$

- We can use Monte Carlo by noting:

$$E[g(U_1, \dots, U_k)] = \int_0^1 \cdots \int_0^1 g(u_1, \dots, u_k) du_1 \cdots du_k$$

for U_i i.i.d. $\sim Unif(0,1)$

Computing Multivariate Integrals

- Similar ideas as the one-dimensional case to address other integration limits can be used to handle

$$\int_{a_k}^{b_k} \cdots \int_{a_1}^{b_1} g(x_1, \dots, x_k) dx_1 \cdots dx_k$$

- That is, by using independent $Unif(a_i, b_i)$ or a change of variables

Computing Multivariate Integrals

- Similarly for

$$\int_0^\infty \cdots \int_{a_1}^{b_1} g(x_1, \dots, x_k) dx_1 \cdots dx_k$$

- That is, by identifying $g(x_1, \dots, x_k) = h(x_1, \dots, x_k)f(x_1, \dots, x_k)$, or using independent change of variables $y_k = \frac{1}{1+x_k}$ for the needed variables

Computing Multiple Integrals

Estimate $\int_0^1 \int_0^1 e^{(x+y)^2} dx dy$

Computing Multiple Integrals

Estimate the area of a unit circle, i.e.,

$$\int_{-1}^1 \int_{-1}^1 I(x^2 + y^2 \leq 1) dx dy$$